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Nitte (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester B.Tech (CBCS) Degree Examinations, December 2025

25EEE101– ELECTRICAL CIRCUIT ANALYSIS - I

(For ACT, ECE, EEE, VDT)

Duration: 3 Hours]

[Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

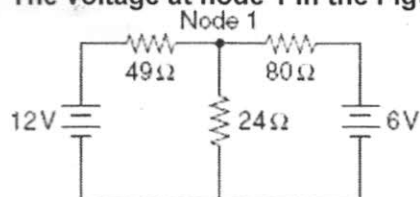
Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

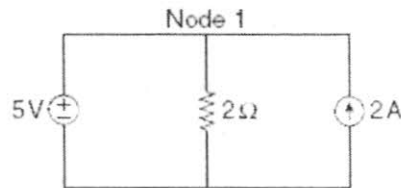
CO1: Q1 to Q4; CO2: Q5 to Q8; CO3: Q9 to Q12; CO4: Q13 to Q16; CO5: Q17 to Q20.

- A network consists of linear resistors and ideal voltage source. If the value of the resistors is doubled then voltage across each resistor is _____
 A) Halved B) Doubled
 C) Increased four times D) Not changed
- Ideal voltage source has _____
 A) Zero internal resistance B) Infinite internal resistance
 C) Low value of current D) Large value of e.m.f.
- A delta circuit has each element of value $R/2$. The equivalent elements of the star circuit will be _____
 A) $R/6$ B) $R/3$
 C) $2R$ D) $3R$
- _____ is an active element
 A) Resistor B) Transistor
 C) Capacitor D) Inductor
- Which of the following is not an expression power?
 A) $P=VI$ B) $P=I^2R$
 C) $P=V^2/R$ D) $P=I/R$
- KVL works on the principle of which of the following _____
 A) law of conservation of charge. B) law of conservation of energy
 C) both. D) None of the above
- The voltage at node 1 in the Figure is _____



- A) 6.15 V B) 12.5 V
 C) 4.25 V D) 3.75 V

8. The voltage at node 1 in the figure below is _____



- A) 2.5 V
C) 3 V
B) 5 V
D) 2 V

9. Peak factor of an AC quantity is defined as _____

- A) Ratio of rms value of ac quantity to its peak value
C) Ratio of peak value of ac quantity to its rms value
B) Ratio of peak value of ac quantity to its average value
D) Ratio of rms value of ac quantity to its average value

10. The power factor is defined as _____

- A) Ratio of Active power to Reactive power
C) Ratio of Apparent power to Active power
B) Ratio of Active power to Apparent power
D) Ratio of Reactive power to Apparent power

11. The voltage applied to the circuit is $v = 10 \sin \left(314.14t + \frac{\pi}{2} \right)$ and current flowing in the circuit is $i = 10 \sin \left(314.14t - \frac{\pi}{4} \right)$. Then the operating power factor is _____

- A) 0.5
C) 0
B) 1
D) 0.707

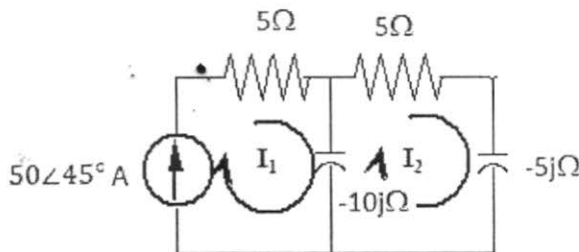
12. A 230V, 50Hz AC supply is applied to a RLC series circuit having $R=10\Omega$, and $X_L=15\Omega$, $X_C=5\Omega$ then the impedance and power factor of the circuit are _____

- A) 1.414, 0.707 lag
C) 14.14, 0.707 lead
B) 14.14, 0.707 lag
D) 1.414, 0.707 lead

13. Kirchhoff law is applicable for _____ domains?

- A) Frequency
C) Coordinate
B) Time
D) Both A and B

14. The current I_1 flowing in the circuit is _____



- A) $50\angle 45^\circ$
C) $10\angle 45^\circ$
B) $50\angle 90^\circ$
D) $10\angle 90^\circ$

15. The quality factor (Q) of a parallel resonant circuit is defined as _____

- A) Ratio of resonant frequency to bandwidth.
C) Ratio of reactance to resistance.
B) Ratio of bandwidth to resonant frequency.
D) Ratio of resistance to reactance.

16. A series RLC circuit with $R = 102$, $L = 1$ mH and $C = 1000$ picofarads is connected across a sinusoidal source of 20V with variable frequency. The resonant frequency of the circuit _____

- A) 159 Hz
C) 159 KHz
B) 100 KHz
D) 100 Hz

- c) Using Source transformation method, find the current flow through the 50V source for the circuit shown in Fig Q 2c

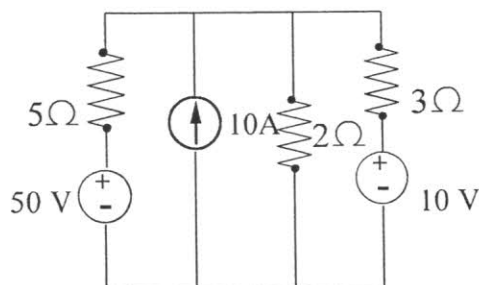


Fig Q 2c

6 L3 1 1,2,3,7,8

Unit – II

3. a) Define and explain Kirchoff's laws with examples
 b) Define the dependent source and list the different types of dependent source with an example.
 c) Find the current i_1 , i_2 and i_3 for the circuit shown in Fig Q 3c using mesh analysis.

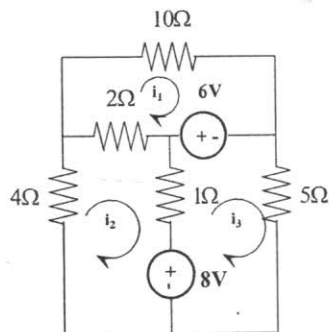


Fig. Q. 3(c)

6 L3 2 1,2,3,7,8

4. a) With an example, explain the super node concept as applied to electric circuit analysis.
 b) Identify the type of dependent sources in the circuit shown in Fig Q 4(b) and hence determine the node voltages of the circuit using node analysis. All resistances are in Ω .

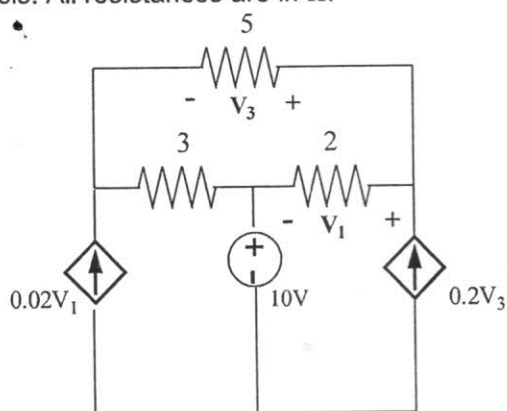


Fig Q 4(b)

6 L3 2 1,2,3,7,8

- c) Find the current 'i' in Fig Q 4(c) by mesh analysis and verify the same by node analysis.

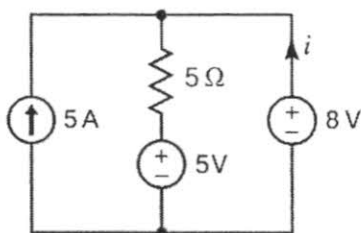


Fig Q 4(c)

Unit – III

- | | | | | | |
|-------|---|---|----|---|-----------|
| 5. a) | Define the following | 6 | L3 | 2 | 1,2,3,7,8 |
| | i. Power Factor | | | | |
| | ii. RMS value of voltage | | | | |
| | iii. Average value of current | | | | |
| | iv. Form factor of a ac voltage | | | | |
| b) | Define average value of AC current and hence derive an expression for average value of a fully rectified sine wave | 4 | L1 | 3 | 1,2,3,7,8 |
| c) | A capacitor of $16\mu\text{F}$ takes a current of 2 A when the alternating voltage applied across it is 230 V. Calculate: | 6 | L2 | 3 | 1,2,3,7,8 |
| | i. the frequency of the applied voltage | | | | |
| | ii. the resistance to be connected in series with the capacitor to reduce the current in the circuit to 1 A at the same frequency | | | | |
| | iii. the phase angle of the resulting circuit. | 6 | L3 | 3 | 1,2,3,7,8 |
| 6. a) | Explain the power triangle in AC circuits | 4 | L2 | 3 | 1,2,3,7,8 |
| b) | Explain the voltage, current and average power waveforms in RL series circuits with relevant equations | 6 | L2 | 3 | 1,2,3,7,8 |
| c) | A 240 V, 50 Hz AC supply is applied a coil of 0.08 H inductance and 4Ω resistance connected in series with a capacitor of $8\mu\text{F}$. Calculate the following | | | | |
| | i. Impedance | | | | |
| | ii. Circuit current | | | | |
| | iii. Phase angle between voltage and current, | | | | |
| | iv. Power factor | | | | |
| | v. Power consumed | | | | |
| | vi. Reactive Power | 6 | L3 | 3 | 1,2,3,7,8 |

Unit – IV

- | | | | | | |
|-------|---|---|----|---|-----------|
| 7. a) | Define resonance in a series RLC circuit and hence derive the expression for resonant frequency | 4 | L2 | 4 | 1,2,3,7,8 |
| b) | For the parallel network shown in the Fig Q 7b below, determine the value of R for resonance | | | | |

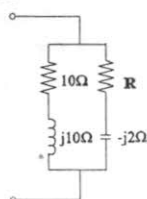


Fig Q 7b

5	L3	4	1,2,3,7,8
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- c) Determine current I_1 and voltage V_1 for the network shown in Fig Q 7c using mesh analysis

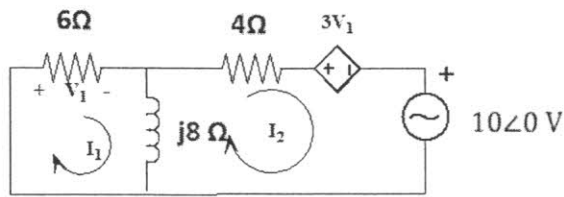


Fig Q 7c

7* L3 4 1,2,3,7,8

8. a) List any four characteristics of parallel resonant circuit
 b) A series RLC circuit has $R=1k\ \Omega$, an inductor of 100mH and capacitor of $10\ \mu F$. An ac supply of 100 V at variable frequency is applied to it. Determine
 i. The resonant frequency
 ii. The resonant current
 iii. The inductive reactance and capacitive reactance at resonance
 iv. The bandwidth of the circuit
 v. The Quality factor Q
 c) Determine the real power output of the source for the circuit shown in Fig Q 8c by nodal analysis. Hence verify KCL at node X.

4 L1 4 1,2,3,7,8

5 L3 4 1,2,3,7,8

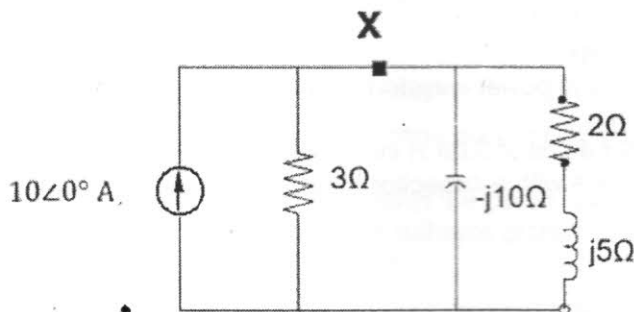


Fig Q 8c

7 L3 4 1,2,3,7,8

Unit – V

9. a) Define the following terms in Magnetic circuits
 i. Magnetic field
 ii. Magnetic Field Strength
 iii. Magnetomotive Force (MMF)
 iv. Magnetic Flux
 b) Derive the expression for induced emf in an single phase transformer and hence calculate the rms value of induced emf in secondary coil of a transformer for a given a sinusoidal flux 0.2 Wb (max.) links with 55 turns of a transformer secondary coil. Assume the supply frequency as 50 Hz

4 L1 5 1,2,3,7,8

6 L2 5 1,2,3,7,8

- c) Calculate the phasor currents I_1 and I_2 in the circuit of Fig Q 9c

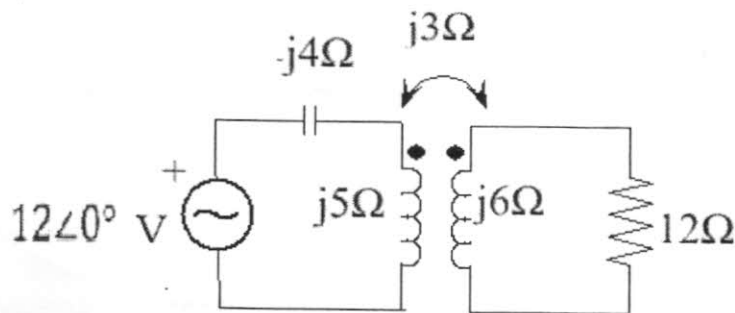


Fig Q 9(c)

10. a) State and explain Faraday's laws of Electromagnetic Induction
 b) With a neat diagram, explain the different types of DC motors
 c) An iron ring of cross-sectional area 6 cm^2 is wound with a wire of 100 turns and has a saw cut of 2 mm as in Fig Q 10c. Calculate the magnetizing current required to produce a flux of 0.1 mWb if mean length of magnetic path is 30 cm and relative permeability of iron is 470.

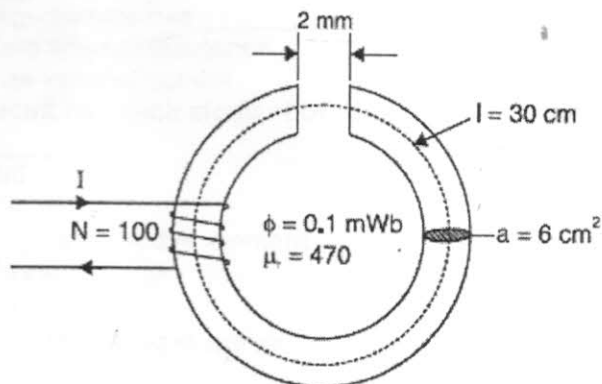


Fig Q 10c.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
